Technology Stack

Date	01 October 2026
Team ID	SWTID-2026-6912
Project Name	FitBuddy - AI Fitness Plan Generator using Gemini Models
Maximum Marks	2 Marks

Define Your Technology Stack:

FitBuddy is a web-based application that generates personalized 7-day workout plans and nutrition or recovery tips using Google Gemini models. The project document specifies FastAPI for backend logic, HTML/CSS with Jinja2 for the frontend, SQLite with SQLAlchemy for data persistence, and Uvicorn for local serving.

The following stack is derived from the FitBuddy project architecture and workflow. Each technology is mapped to the role it performs in the application.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side** User interface and dynamic pages	**HTML + CSS + Jinja2**	Provides the user input form, personalized result page and admin page. Jinja2 dynamically renders backend data into HTML templates, while CSS supports responsive layouts and consistent styling.
2	**Backend / Server-Side** Application logic and routing	**Python + FastAPI**	Handles routing, form processing, validation, API endpoints and coordination between the frontend, Gemini services and database. FastAPI is the documented backend framework for FitBuddy.
3	**AI / Generative Layer** Workout, feedback and nutrition generation	**Google Gemini 1.5 Pro + Gemini Flash**	Gemini 1.5 Pro generates structured 7-day workout plans and updates them from feedback. Gemini Flash generates fast nutrition or recovery tips.
4	**Database / Data Storage** Persistent user and plan data	**SQLite + SQLAlchemy**	Stores user information and AI-generated plans. SQLAlchemy ORM provides the database layer, while SQLite supplies local persistent storage.
5	**Cloud / Hosting / Deployment** Application execution environment	**Uvicorn + Local Deployment**	Runs the FastAPI application locally for development and verification. The project document identifies local deployment as the current deployment milestone and prepares the application for future cloud deployment.

FitBuddy – Team Work Assignment

The work is assigned according to the order of names provided. Each member is mapped to a major project area from the documented FitBuddy workflow, while the Team Leader coordinates integration and final review.

S.No	Name / Role	Assigned Work	Related Project Activities
1	**Pavithra M** Team Leader	**Project coordination + Model & Architecture** Coordinate the overall project, finalize architecture, coordinate Gemini model selection, maintain task integration and conduct final review.	Milestone 1: Model selection and Architecture Activities 1.1, 1.2, 1.3
2	**Manisha S** Team Member	**Core AI functionality** Develop/maintain workout-plan generation, nutrition/recovery tip generation and feedback-based plan updating using Gemini modules.	Milestone 2: Core functionalities Development Activity 2.1
3	**Monica G** Team Member	**FastAPI backend + database integration** Handle form processing, validation, routing support and storage of users and plans using FastAPI, SQLite and SQLAlchemy.	Milestone 2: Activity 2.2 Database functions and backend integration
4	**Monisha Anandan** Team Member	**routes.py / application logic** Implement and maintain the main operational routes connecting frontend forms, Gemini functions and database operations.	Milestone 3: routes.py Development Activity 3.1
5	**Mridula M** Team Member	**Frontend + templates + deployment verification** Design responsive HTML/CSS pages, Jinja2 templates, UI flow and assist with local deployment testing and verification.	Milestone 4: Frontend Development Activities 4.1, 4.2 Milestone 5: Deployment Activities 5.1, 5.2

Project Workflow Reference

1. Model selection and Architecture → 2. Core functionalities Development → 3. routes.py Development → 4. Frontend Development → 5. Deployment

The assignments above follow this documented sequence so that responsibilities progress from architecture and AI selection through backend, routing, frontend, and deployment verification.

Technology Stack Summary

Layer	Selected Technology
Frontend	HTML, CSS, Jinja2
Backend	Python, FastAPI
AI	Google Gemini 1.5 Pro, Gemini Flash
Database	SQLite, SQLAlchemy
Server / Deployment	Uvicorn, Local Deployment